International Trade and Heterogeneous Firms

ECON 404: Cal Poly

Introduction

- Why do we need firm heterogeneity?
- Krugman model has gains from variety, but extensive margin was agnostic.
- Better understanding of this extensive margin leads to a better understanding of the world.
- It's cool.

Road Map

1. Consumers demand varieties of a good.
2. Entrepreneur decides to pay a cost to take a draw from a productivity distribution.
3. Entrepreneur observes productivity and decides to become a firm.
4. Non-negative profit condition.
5. Free-entry to close the model.
6. Comparative Statics.

Consumers

- First define $\varepsilon = 1/(1-\alpha) > 1$ is the elasticity of substitution and Ω is the set of varieties available to the consumer.
- Utility

$$U = \mu \ln(X) + Y$$

$$X = \left(\int_{i \in \Omega} x(i)^{\alpha} di \right)^{\frac{1}{\alpha}}$$

- Demand

$$x(i) = \left[\frac{p(i)^{-\varepsilon}}{\mathcal{P}^{1-\varepsilon}} \right] \mu$$

where

$$\mathcal{P}^{1-\varepsilon} = \int_{i \in \Omega} p(i)^{1-\varepsilon} di$$

Firms

- Before thinking about entrepreneurs, we need a brief discussion on labor.
- Labor is fixed, fully employed, and only input.
- Since labor is only input, total income $= w * L = E$.
- Numeriare is freely traded, equalizes wages, and normalize wage $= 1$.

Entrepreneurs

- Pay a fixed cost f_E (measured in units of labor) in order to enter the industry.
- Draw a constant marginal cost a from the distribution $G(a)$.
- Can pay a fixed cost f_D to set up shop and serve the domestic market.
- Can pay another fixed cost $f_X > f_D$ to serve the foreign market through exports.
    - If I export, I pay a symmetric iceberg transport cost represented by $\tau > 1$.
- Can exit and do nothing.

Firms

- Note that with constant marginal costs the decisions to serve the domestic and foreign market are independent.
- Do I pay f_D?
  - Only if my variable profit is at least as big as f_D.
- Domestic profits are

$$\begin{aligned} \pi_D(i) &= \left[p(i) - a_i\right] Q_D(i) - f_D \\ &= \left[\frac{[p(i) - a_i]\mu}{\mathcal{P}^{1-\varepsilon}}\right] p(i)^{-\varepsilon} - f_D. \end{aligned}$$

- Note $Q_D(i) = x(i)$ in equilibrium; i.e. supply $=$ demand.

Firms

- A firm selling domestically will charge a price equal to a constant markup over marginal cost,

$$p(i) = \frac{a_i}{\alpha}.$$

- Therefore, the operating profit function for a purely domestic firm is

$$\pi_D(i) = a_i^{1-\varepsilon} B - f_D$$

where

$$B = \frac{1}{\varepsilon \alpha^{1-\varepsilon}} \left(\frac{\mu}{\mathcal{P}^{1-\varepsilon}} \right).$$

Equilibrium Condition 1

- Define a_D to be the least efficient firm that sells to the domestic market.
- This firm is given by

$$a_D^{1-\varepsilon} B = f_D.$$

- This means that every entrepreneur that drew an $a < a_D$ will produce and serve the domestic market.
- Any entrepreneur that drew an $a > a_D$ will not produce.

- Note that there is the analogous condition for the foreign country: ${a_D^*}^{1-\varepsilon} B^* = f_D$

Firms

- Do I pay f_X?
  - Only if my variable profit from exports is at least as big as f_X.
- Export profits are

$$\pi_X = (\tau a_i)^{1-\varepsilon} B^* - f_X,$$

where

$$B^* = \frac{1}{\varepsilon \alpha^{1-\varepsilon}} \left(\frac{\mu}{\mathcal{P}^{*1-\varepsilon}} \right).$$

- Note again that price is a constant markup of marginal cost $p^*(i) = \frac{\tau a_i}{\alpha}$ and export supply equal foreign demand.

Equilibrium Condition 2

- Define a_X to be the least efficient firm that sells to the foreign market.
- This firm is given by

$$(\tau a_X)^{1-\varepsilon} B^* = f_X.$$

- This means that every entrepreneur that drew an $a < a_X$ will produce and serve the foreign and domestic market.
- Any entrepreneur that drew an $a_X < a$ will not export.
- Note that there is the analogous condition for the foreign country: $\tau {a_X^*}^{1-\varepsilon} B = f_X$

Free Entry

- We have characterized the cutoff firms, but we aren't done.
- First notice that

$$B = \frac{1}{\varepsilon \alpha^{1-\varepsilon}} \left(\frac{\mu}{\mathcal{P}^{1-\varepsilon}} \right)$$

and

$$\begin{aligned} \mathcal{P}^{1-\varepsilon} &= \int_{i \in \Omega} p(i)^{1-\varepsilon} di \\ &= N_E \int_0^{a_D} \left(\frac{a}{\alpha} \right)^{1-\varepsilon} dG(a) + N_E^* \int_0^{a_X^*} \left(\frac{\tau a}{\alpha} \right)^{1-\varepsilon} dG(a) \end{aligned}$$

- So, we need to figure out the number of people taking a draw in each country, N_E and N_E^*

Free Entry

- The free entry condition implies that expected profits $\bar{\pi}$ must be zero.
- I.e. the probability I draw a productivity good enough to sell domestically and the associated payoff plus the probability I draw a productivity good enough to be an exported and the associated payoff must equal the cost to take a draw.

$$\left[V(a_D)B - G(a_D)f_D\right] + \left[V(a_X)\tau^{1-\varepsilon}B^* - G(a_X)f_X\right] = f_E$$

where

$$V(z) = \int_0^z a^{1-\varepsilon} dG(a).$$

Distribution Assumption

- Though we don't need to have a specific distribution to derive the main results, it is nice to have closed form solutions.
- It is common (and matches US data well) to assume $G(a)$ follows the Pareto distribution:

$$G(a) = \left(\frac{a}{a_U}\right)^k, \; 0 < a < a_U$$

$$V(a) = \frac{k}{k+1-\varepsilon}\left(\frac{a}{a_U}\right)^k a^{1-\varepsilon}, \; 0 < a < a_U$$

  where $k > (\varepsilon - 1)$.
- Side Note: Some researchers are also using the Fréchet distribution or truncated Pareto.

Equilibrium Condition 3

- Using the Pareto distribution, our free entry condition becomes

$$f_E = \frac{(\varepsilon - 1)[a_D^k f_D + a_X^k f_X]}{[k + 1 - \varepsilon]a_U^k}.$$

- Similarly, for the foreign country

$$f_E = \frac{(\varepsilon - 1)[{a_D^*}^k f_D + {a_X^*}^k f_X]}{[k + 1 - \varepsilon]a_U^k}$$

Recap

- We have a total of six equilibrium conditions – 3 conditions $\times$ 2 countries.
  1. Nonnegative profit for domestic market.
  2. Nonnegative profit for foreign market.
  3. Free entry.

- Given our assumptions, we will have a symmetric equilibrium

$$a_D = a_D^*, \qquad a_X = a_X^*, \qquad N_E = N_E^*$$

- So we really only have three equilibrium conditions now.

Solution

$$a_A = \left[\frac{\psi f_E}{f_D}\right]^{\frac{1}{k}} a_U,$$

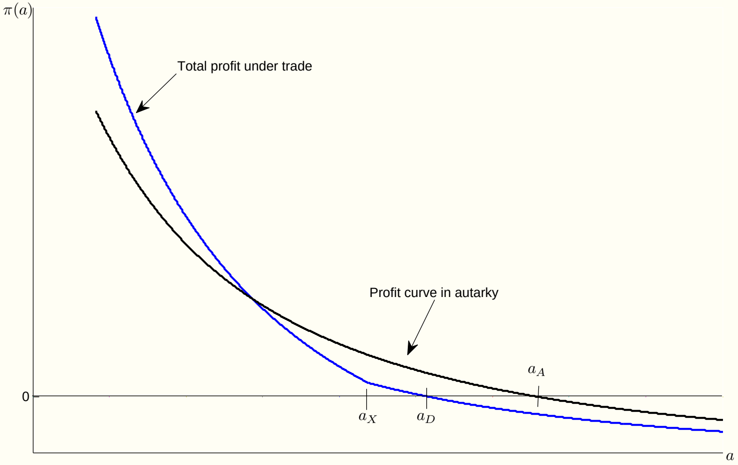
$$a_D = \left[\frac{f_X^{\psi}\tau^k}{f_X^{\psi}\tau^k + f_D^{\psi}} \frac{\psi f_E}{f_D}\right]^{\frac{1}{k}} a_U,$$

$$a_X = \left[\frac{f_D^{\psi}}{f_X^{\psi}\tau^k + f_D^{\psi}} \frac{\psi f_E}{f_X}\right]^{\frac{1}{k}} a_U,$$

$$N_E = \frac{\alpha\mu}{k f_E}$$

where

$$\psi \equiv \frac{[k - (\varepsilon - 1)]}{(\varepsilon - 1)} > 0.$$



$\pi(a)$
Total profit under trade
Profit curve in autarky
a_A
0
a_X
a_D
a

Results

- Opening up to trade causes the least efficient firms to exit

$$-\frac{da_D}{d\tau} = \left(\frac{-f_D^{\psi}}{f_X^{\psi}\tau^k + f_D^{\psi}} \right) \frac{a_D}{\tau} < 0$$

- Note that this can easily be expressed as an elasticity

$$-\frac{\tau}{a_D}\frac{da_D}{d\tau} = \left(\frac{-f_D^{\psi}}{f_X^{\psi}\tau^k + f_D^{\psi}} \right) \equiv -\epsilon_{a_d}$$

Results

- Opening up to trade increases the number (mass) of varieties imported/exported

$$-\frac{da_X}{d\tau} = \left(\frac{f_X^{\psi} \tau^k}{f_X^{\psi} \tau^k + f_D^{\psi}} \right) \frac{a_X}{\tau} > 0$$

- Note that this can easily be expressed as an elasticity

$$-\frac{\tau}{a_X} \frac{da_X}{d\tau} = \left(\frac{f_X^{\psi} \tau^k}{f_X^{\psi} \tau^k + f_D^{\psi}} \right) \equiv -\epsilon_{a_X}$$

Results

- What about total variety gains à la Krugman?
- The mass of varieties in each group are

$$N_D = G(a_D)N_E = \left(\frac{a_D}{a_U}\right)^k N_E$$

$$N_X = G(a_X)N_E = \left(\frac{a_X}{a_U}\right)^k N_E.$$

- So, the change in total available varieties to consume is

$$-\frac{dN_C}{d\tau} = -\left[\frac{k a_D^{k-1} N_E}{a_U^k}\frac{da_D}{d\tau} + \frac{k a_X^{k-1} N_E}{a_U^k}\frac{da_X}{d\tau}\right]$$

$$= \frac{\psi\alpha\mu}{f_X \tau f_D}\left[\frac{f_X^{\psi}\tau^k f_D^{\psi}[f_D - f_X]}{(f_X^{\psi}\tau^k + f_D^{\psi})^2}\right] < 0$$

Results

- So if there are no gains from variety, where do they come from?
- (Note that the variety result is not general to all distributions and models)
- Efficiency gains.
- Through international competition, the least efficient firms exit and are no longer paying fixed costs.
- What about consumers?
- Though they lose some varieties, the varieties that are left are less expensive

Welfare

- To see this more concretely, look at the welfare function.
- Welfare is the indirect utility.

$$V = \mu \ln(X) + (I - \mu). \tag{1}$$

- Income/Expenditure is equal to labor income plus profits from domestically owned firms:

$$I = L_k + N_E \bar{\pi}.$$

- What are average aggregate profits?

Welfare

- Dixit and Stiglitz (1977) show that

$$X = \frac{Is(\mathcal{P})}{\mathcal{P}}$$

where $s(\mathcal{P})$ is the propensity to consume the heterogeneous good. Quasilinear utility implies that $Is(\mathcal{P}) = \mu$. Thus

$$\begin{aligned} \mu &= \mathcal{P}X \\ \Rightarrow 0 &= X\frac{d\mathcal{P}}{d\tau} + \mathcal{P}\frac{dX}{d\tau} \\ \Rightarrow \frac{1}{X}\frac{dX}{d\tau} &= \frac{-1}{\mathcal{P}}\frac{d\mathcal{P}}{d\tau} \end{aligned}$$

Welfare

- Differentiating the (negative) indirect utility (1), we get

$$
\begin{aligned}
-\frac{dV}{d\tau} &= -\frac{\mu}{X}\frac{dX}{d\tau} \\
&= \frac{\mu}{\mathcal{P}}\frac{d\mathcal{P}}{d\tau}
\end{aligned}
$$

- So the change in welfare depends on how the price index changes.
- The price index is a weighted average of prices and depends on
  1. Prices
  2. Number of varieties

Price Index

- We can write the price index in the following way

$$\mathcal{P} = \left(\left[\frac{k}{k+1-\varepsilon} \right] \left[N_D \left(\frac{a_D}{\alpha} \right)^{1-\varepsilon} + N_X \left(\frac{\tau a_X}{\alpha} \right)^{1-\varepsilon} \right] \right)^{\frac{1}{1-\varepsilon}}$$

- Important things to note
  - For welfare $\uparrow$, we need $\tau \downarrow \Rightarrow \mathcal{P} \downarrow$, and
  - $\varepsilon > 1$
- I.e. $\tau \downarrow \Rightarrow (\mathit{stuff}) \uparrow$

$$\mathcal{P} = \frac{1}{\alpha} \left(\left[\frac{k N_E}{k+1-\varepsilon} \right] \left[\frac{a_D^{k+1-\varepsilon} + \tau^{1-\varepsilon} a_X^{k+1-\varepsilon}}{a_U^k} \right] \right)^{\frac{1}{1-\varepsilon}}$$

Price Index

- Differentiating the price index yields the following elasticity

$$\frac{\tau}{\mathcal{P}}\frac{d\mathcal{P}}{d\tau} = \left[\frac{-\psi}{\alpha}\right]\left[\frac{a_D^{k+1-\varepsilon}\epsilon_{a_D} + \tau^{1-\varepsilon}a_X^{k+1-\varepsilon}\epsilon_{a_X} - \left(\frac{1}{\psi}\right)\tau^{1-\varepsilon}a_X^{k+1-\varepsilon}}{a_D^{k+1-\varepsilon} + \tau^{1-\varepsilon}a_X^{k+1-\varepsilon}}\right]$$

- Note that $\frac{\tau a_X}{a_D} = \left(\frac{f_D}{f_X}\right)^{\frac{1}{\varepsilon-1}}$

$$\frac{\tau}{\mathcal{P}}\frac{d\mathcal{P}}{d\tau} = \left[\frac{-\psi}{\alpha}\right]\left[\frac{f_X^{\psi}\tau^k\epsilon_{a_D} + f_D^{\psi}\epsilon_{a_X} - \left(\frac{f_D^{\psi}}{\psi}\right)}{f_X^{\psi}\tau^k + f_D^{\psi}}\right]$$

Welfare

- Using our elasticities, this collapses to

$$\frac{\tau}{\mathcal{P}}\frac{d\mathcal{P}}{d\tau} = \left[\frac{1}{\alpha}\right]\left[\frac{f_D^{\psi}}{f_X^{\psi}\tau^k + f_D^{\psi}}\right] = \frac{\epsilon_{a_D}}{\alpha} > 0 \tag{2}$$

- Therefore our welfare analysis concludes with

$$-\frac{dV}{d\tau} = \left(\frac{\mu}{\alpha\tau}\right)\epsilon_{a_D} > 0 \tag{3}$$

So who cares?

- First, there are some important results from the Melitz model:
  - Even without gains from variety, countries can gain from trade though better efficiency.
  - The most efficient (and biggest) are the firms who export (which has been the source of numerous empirical research)
- But perhaps more importantly, this has given us a tractable model to allow for more realistic extensive margin...
- which can have important implications for models from the past (Krugman vs. Chaney), but also give us the tools to ask really interesting questions.

Some applications of the "Meltiz" model

Cole and Davies (2014 RDE)

- Really thinking about the variety effects on welfare; I.e. Does trade lead to homogenization/Americanization?
- Utility

$$U = \mu_1 \ln (X_1) + \mu_2 \ln (X_2) + \Phi(a_D, a_D^*) + Y$$

where

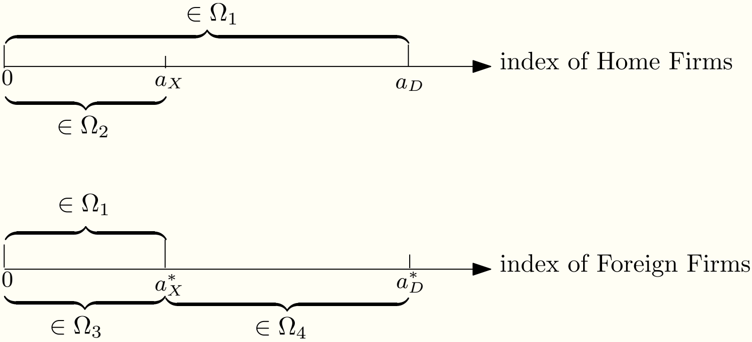
$$X_1 = \left(\int_{i \in \Omega_1} x_k(i)^\rho di \right)^{\frac{1}{\rho}},$$

$$X_2 = \left[\alpha^{\frac{1}{\varepsilon}} \left(\int_{i \in \Omega_2} x(i)^\rho di \right) + \beta^{\frac{1}{\varepsilon}} \left(\int_{i \in \Omega_3} x(i)^\rho di \right) + \gamma^{\frac{1}{\varepsilon}} \left(\int_{i \in \Omega_4} x(i)^\rho di \right) \right]^{\frac{1}{\rho}}$$

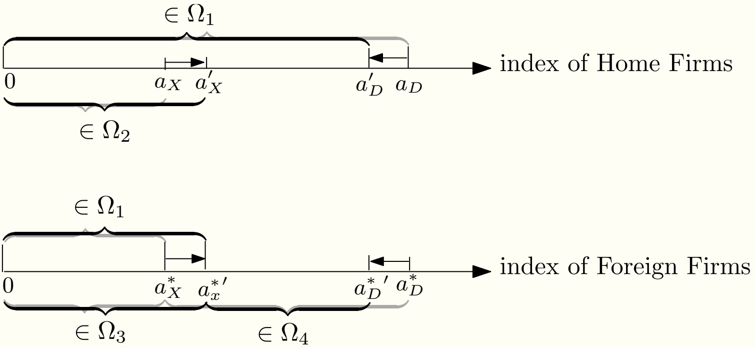
and

$$\mu_1 > 0, \ \mu_2 \geq 0$$

Equilibrium Cutoffs



Changes in Cutoffs



Cole and Davies (2014 RDE) – Conclusion

- Goal was to take the critique of variety homogenization and give it full force
  - Small, local, unusual varieties are especially prized
  - Potential loss from globalization
- Find that a reduction in trade costs increases welfare from direct consumption
  - An ambiguous effect on tourism benefit
  - Total effect still positive since choice of exporting is endogenous
- Only the loss in the existence value can lead to a welfare loss

Helpman, Melitz, and Yeaple (2004 AER)

- Introduced the option to become a multinational in addition to exporting.
- Everything very similar except firms could now pay a fixed cost $f_M > f_X > f_D$ to participate in FDI.
- This results in the most productive firms being MNEs, the "medium" productive firms exporting, and the least productive firms selling only to the domestic market.
- This is confirmed in the data and they find that the productivity advantage of MNEs over exporters is about 15% for US firms.

Cole and Davies (2011 EER)

- Investigates how the presence or threat of FDI affects a country's "optimal" tariff.
- Ellingsen and Wärneryd (1997 IER) have homogeneous firms and show how the "optimal" tariff (one that maximizes domestic firm profits) is one that is just low enough to prevent firms from tariff-jumping.
- Firm heterogeneity dulls this knife-edge result and allows for some firms to be MNEs while others export.
- They assume firms are heterogenous over fixed cost instead of marginal cost and don't allow for drawing productivity from a distribution.
- They find that the presence of FDI lowers the noncooperative Nash tariff.

Midterm Review

- Exam:
  - About 6-7 questions: multiple parts
  - Can have a “clean” calculator, pencil, and a straight-edge
  - Answers in provided space – scratch paper (not turned in) will be provided
- Ricardian Model
  - Labor only input
  - Comparative advantage/oportunity cost
  - Everyone gains
  - In general leads to full specialization
  - Except when countries are really different in size.

Midterm Review

- Specific Factors model
  - Labor fully mobile, but other factors specific to an industry is immobile
  - PPF bowed out and slope is equal to $-\frac{MPL_F}{MPL_C}$
  - How do you find wages?
  - In general:
    1. The factor specific to the sector whose relative price increases is definitely better off.
    2. The factor specific to the sector whose relative price decreases is definitely worse off.
    3. The change in welfare for the mobile factor is ambiguous.
  - Begin to think about political economy of trade

Midterm Review

- Hechscher-Ohlin model
  - Standard model assumes identical technologies, but different endowments of capital and labor.
  - Capital and Labor substitutable – bowed out ppf with slope equal to $-\frac{MPL_F}{MPL_C} = -\frac{MPK_F}{MPK_C}$
  - how to find out Capital and Labor intensive industry
  - **The H-O Theorem:** A country will have comparative advantage in, and therefore will export, that good whose production is relatively intensive in the factor with which that country is relatively well endowed.
  - **The Rybczynski Theorem:** At constant world prices, if a country experiences an increase in the supply of one factor, it will produce more of the product whose production is intensive in that factor and less of the other.

Midterm Review

- Hechscher-Ohlin model
  - **The Factor Price Equalization Theorem:** Given all the assumptions of the H-O model, free international trade will lead to the international equalization of individual factor prices.
  - **The Stolper-Samuelson Theorem:** Free international trade benefits the abundant factor and harms the scarce factor.
  - **Leotief Paradox**
  - Empirical evidence of H-O Theorem.

Midterm Review

- Distributional Effects of Trade
  - Autor, Dorn and Hanson (2013 AER) – Effect of China shock
  - Pierce and Schott (2016 AER) – Effect of China entering the WTO
  - McLaren and Hakobyan (2016 REStat) – Effect of NAFTA
  - Main takeway is non-college educated workers in particular in manufacturing (but also in non-manufacturing sectors) made worse off in terms of employment and wages. College educated workers saw little to no effect.

Midterm Review

- New Trade Theory
  - Intra-industry trade
  - Can have everything identical (but also have increasing returns to scale; e.g. add a fixed cost to production and constant marginal cost) and gain from trade through gains in variety.
- New New Trade Theory
  - New Trade Theory, but with firm heterogeneity.
  - Now we don't necessarily gain from trade through increased variety, but we do gain from increased efficiency by "killing off" the least productive firms.